

Name _____

Date _____

Total
/41

KNOWLEDGE

1. For the following function, $f(x) = x^4 - 2x^2 + 3$, determine the following:
 - a) Find the intervals on which $f(x)$ is increasing and decreasing.
 - b) Find the local maximum and minimum values of $f(x)$.
 - c) Find the intervals of concavity and the inflection points.

(6)

2. For the given function, $f(x) = \sin x$, $a = \frac{\pi}{6}$

- a. Find the Linearization $L(x)$ of the function at a .

(2)

- b. How accurate is the approximation $L(a + 0.1) \approx f(a + 0.1)$? **Use all decimal values.**

(1)

3. a. Find the differential dy and b. evaluate dy for the given values of x and dx

$$y = \sqrt{3 + x^2}, \quad x = 1, \quad dx = -0.1$$

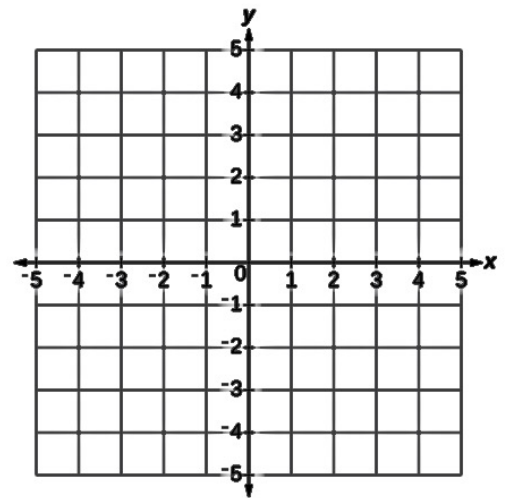
(2)

APPLICATION

4. Sketch the graph of a function that satisfies all the given conditions.

- $f'(x) > 0$ if $|x| < 2$, $f'(x) < 0$ if $|x| > 2$
- $f'(2) = 0$, $\lim_{x \rightarrow \infty} f(x) = 1$, $f(-x) = -f(x)$
- $f''(x) < 0$ if $0 < x < 3$, $f''(x) > 0$ if $x > 3$

(4)



5. If $f(x) = ax^3 + bx^2 + cx + d$, $a \neq 0$, find a , b , c , and d so that $f(x)$ has a local minimum at 0, a local maximum at 4, and the graph contains the points $(0, 5)$ and $(4, 33)$.

(4)

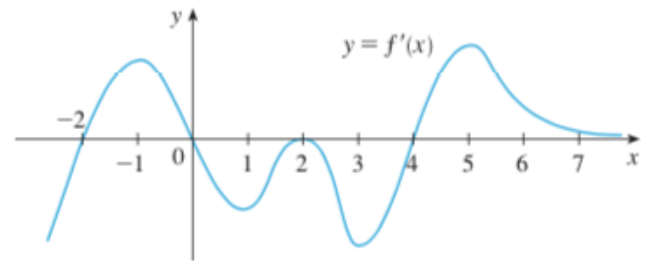
6. Two sides of a triangle are 4 m and 5 m in length and the angle between them is increasing at a rate of 0.06 rad/s. Find the rate at which the area of the triangle is increasing when the angle between the sides of fixed length is $\frac{\pi}{3}$.

(3)

COMMUNICATION

7. Are the following statements about the graph of $f'(x)$ true or false? Be sure to **justify** your response.
- a) The graph of $f(x)$ has a total of 5 inflection points.

(2)



- b) The graph of $f(x)$ has local minimums at $x = -2$ and $x = 0$ and local maximums at $x = 4$ and $x = 2$.

(2)

- c) The graph of $f(x)$ is concave down on the intervals $(-1, 0)$, $(2, 3)$, and $(5, 7)$.

(2)

8. Jeff claimed that the absolute maximum value of the function $f(x) = -x^3 + 6x^2 + 36x + 8$ on the interval $0 \leq x \leq 5$ is 224. Is Jeff's claim correct? Explain.

(3)

THINKING

9. A pair of scuba divers wishes to dive on a wreck that lies 0.4 km off the shore from a point 0.8 km from their present position. If they can walk carrying the equipment at 5 km/h and swim at 3 km/h, how far should they walk along shore before starting to swim in order to reach the wreck in the shortest amount of time? **You do not need to check for a minimum in this question.**

(5)

10. A line through the point $(-4, 1)$ forms a right triangle with the x-axis and y-axis in the 2nd quadrant. Determine the equation of the line that will minimize the area of the triangle.

(5)